

Predicting Chaotic Traffic Flows in Blacksburg, VA

ECE 5694: Nonlinearity & Predictability (Fall 2024)

Ankush Chaudhary

*Department of Electrical Computer Engineering
Virginia Polytechnic Institute and State University
Blacksburg, VA – 24060*

ABSTRACT

Traffic congestion poses a significant challenge to urban mobility, impacting daily commutes and long-term city planning. This project addresses these issues by utilizing Long Short-Term Memory (LSTM) neural networks to predict traffic speed and congestion levels in Blacksburg, Virginia. By integrating real-time data from the APIs like HERE Traffic and OpenWeatherMap, the model offers actionable insights to optimize transportation infrastructure. Through meticulous feature engineering, robust modeling, and precise evaluation, the project showcases its scalability and practical applicability.

1. Introduction

Traffic congestion significantly impacts daily life and planning. Drawing inspiration from Tesla’s dynamic routing algorithms, this project delves into the application of LSTM neural networks to predict traffic conditions in Blacksburg, Virginia. By integrating real-time traffic and weather data through APIs, the study seeks to address urban mobility challenges with precise and timely forecasts. The primary focus is on integrating cutting-edge machine learning techniques to deliver reliable and adaptable solutions for traffic management.

2. Data Collection

timestamp	description	current_speed	free_flow_speed	jam_factor	length	rwf	weather_main	temperature
2024-11-27 19:20:01.079819	VA-412/Prices Fork Rd	8.611112	10.555556	1.4	888.0	0.2578946670360140	Clouds	7.67
2024-11-27 19:20:01.079833	VA-114/Peppers Ferry Rd NE	15.277779	18.61111	1.4	3479.0	0.2507461081042450	Clouds	7.67
2024-11-27 19:20:01.079838	VA-114/Peppers Ferry Rd	29.166668	29.166668	0.0	3344.0	0.0	Clouds	7.67
2024-11-27 19:20:01.079841	US-460	11.666667	13.055556	0.8	33.0	0.08510638688999530	Clouds	7.67

Figure 1: Sample Dataset Structure

This figure provides a glimpse into the dataset used for the project. It includes key details like traffic speed, congestion levels, timestamps, and weather conditions. The image helps set the stage by showing how diverse and detailed the data is.

Over the course of four weeks, the project collected over 80,000 records of real-time traffic and weather data in Blacksburg. These data points were captured at five-minute intervals using two primary APIs:

1. **HERE Traffic API:** This API provides detailed traffic metrics, including current speed, free flow speed, congestion levels (Jam Factor), road descriptions, and timestamps.
2. **OpenWeatherMap API:** This API offers comprehensive weather data, including temperature and descriptive weather conditions such as clear, cloudy, or rainy.

The integration of these APIs ensured the creation of a comprehensive dataset that could be used for model training and validation.

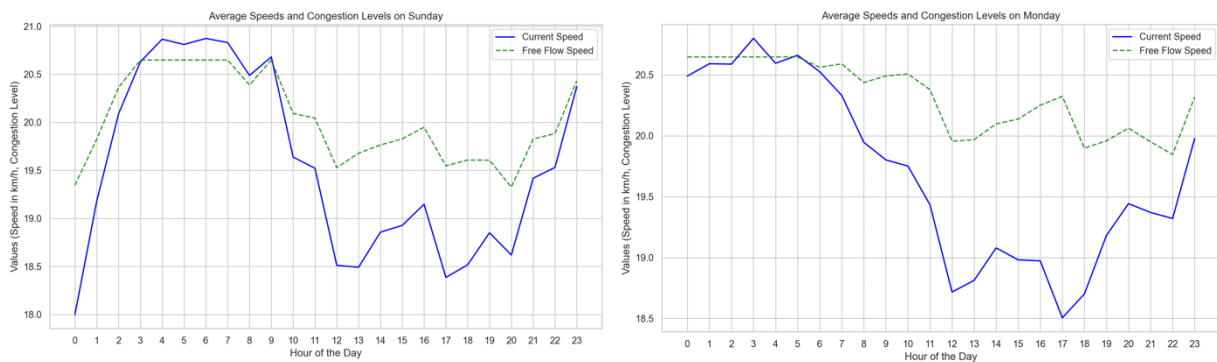


Figure 2: Visualization of Traffic Speed and Jam Factor Distribution

This plot showcases the range and patterns of traffic speeds and congestion levels. It highlights how the data is spread out and identifies any unusual values, giving context to the challenges of predictive modeling.

Features

- **Traffic Data:** This data includes current speed, free flow speed, Jam Factor, road descriptions, road length, and timestamps.
- **Weather Data:** This data captures temperature and descriptive weather conditions that influence traffic patterns.
- **RWF (Roadway Friction Factor):** This critical metric is calculated using the formula:

$$\text{RWF} = \frac{(\text{Free Flow Speed} - \text{Current Speed})}{\text{Free Flow Speed}} \times \text{Jam Factor}$$

This formula quantifies road congestion, enabling the model to predict dynamic traffic conditions more accurately.

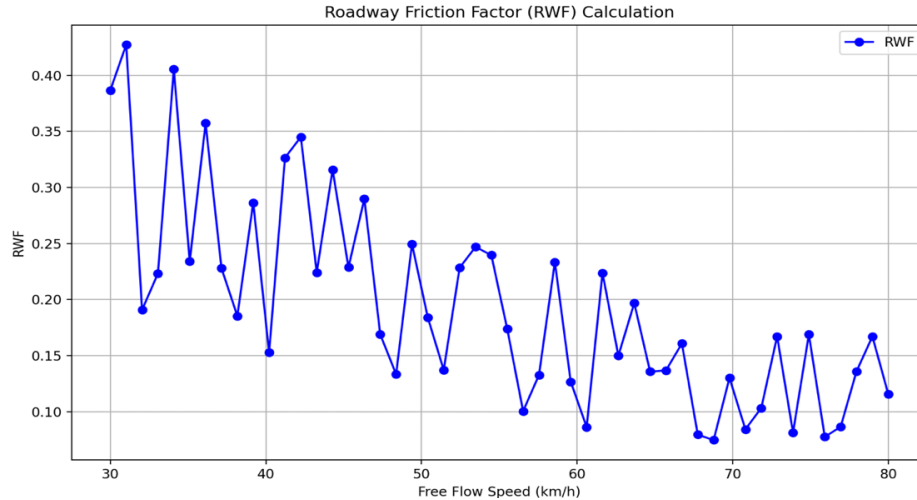


Figure 3: Roadway Friction Factor (RWF) Calculation

This diagram explains how RWF is calculated by combining traffic speed, free flow speed, and congestion factors, this provides a clear picture of road conditions, making it an essential factor.

3. Feature Engineering

Transforming raw data into actionable features was a key component of the project, significantly enhancing the model's performance:

- **One-Hot Encoding:** Converted categorical data, such as weather conditions and road descriptions, into numerical representations to enhance model interpretability.
- **Timestamp Decomposition:** Extracted features like the hour of the day and day of the week to capture temporal patterns in traffic volume.
- **Imputation:** Addressed missing values by replacing numerical gaps with the mean and categorical gaps with the mode.
- **Normalization:** Scaled all features to a $[0, 1]$ range using MinMaxScaler, ensuring efficient model training and convergence.

weather_main_Clear	weather_main_Clouds	weather_main_Drizzle	description_Ellett Rd	description_Roanoke St	description_Toms Creek Rd	hour	day_of_week
0.0	1.0	0.0	0.0	1.0	0.0	19	2
0.0	1.0	0.0	1.0	0.0	0.0	19	2
0.0	1.0	0.0	0.0	0.0	1.0	19	2
0.0	1.0	0.0	0.0	1.0	0.0	19	2
0.0	1.0	0.0	0.0	0.0	0.0	19	2

Figure 4: Feature Engineering Overview

This visualization breaks down the data preparation process, including converting categorical data into numbers, extracting useful time-based patterns, filling in missing values, and scaling the data for better model performance. It highlights how raw data was transformed into meaningful inputs for the model.

4. Model Development

LSTM neural networks were selected due to their ability to handle sequential data, making them suitable for traffic prediction tasks.

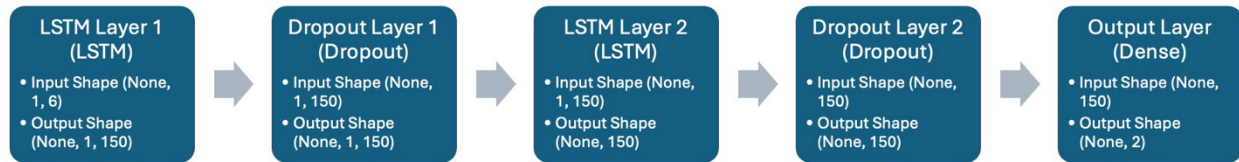


Figure 5: LSTM Model Architecture

This diagram details the structure of the LSTM model used for predictions. It includes layers for processing sequential data, techniques like dropout to prevent overfitting, and an output layer that predicts both traffic speed and congestion levels.

4.1. Architecture

- **Input Layer:** The input layer processes sequential traffic and weather data.
- **Hidden Layers:** The model comprises two LSTM layers, each with 150 units, to prevent overfitting through dropout regularization.
- **Output Layer:** A dense layer predicts traffic speed and congestion levels.
- **Loss Function:** Huber loss is chosen for its robustness against outliers.

4.2. Training Details

- **Early Stopping:** Training is halted when the validation loss fails to improve for 10 consecutive epochs.
- **Evaluation Metrics:** Mean Absolute Error (MAE) and R^2 scores are used to assess prediction accuracy and variance explained by the model.

```
Epoch 97/100
1764/1764 ————— 3s 2ms/step - loss: 3.9003e-04 - val_loss: 2.9789e-04
Epoch 98/100
1764/1764 ————— 3s 2ms/step - loss: 3.8932e-04 - val_loss: 3.0832e-04
Epoch 99/100
1764/1764 ————— 3s 2ms/step - loss: 3.9610e-04 - val_loss: 2.8964e-04
Epoch 100/100
1764/1764 ————— 3s 2ms/step - loss: 3.9134e-04 - val_loss: 3.3477e-04
441/441 ————— 0s 715us/step
```

Figure 6: Training Loss and Validation Loss Progression

This graph tracks how the model's accuracy improved during training. By comparing training loss to validation loss, it demonstrates the model's learning process and shows that overfitting was avoided, thanks to early stopping.

5. Results

Evaluation Metrics:

- **Climatology Baseline:**

- MAE (Speed): 3.11
- MAE (Jam Factor): 0.75
- R^2 (Speed): 0.77

The climatology baseline uses historical data to provide a static reference for comparison. It calculates the aggregated mean of traffic data segmented by road descriptions, hours, and days of the week to establish a benchmark.

- **LSTM Model:**

- MAE (Speed): 0.28
- MAE (Jam Factor): 0.09
- R^2 (Speed): 0.99

The LSTM model processes dynamic features like weather conditions, road friction factors, and temporal trends to produce highly accurate predictions. By incorporating these inputs, the LSTM model outperforms the static climatology baseline.

- **IGN Scores**

- Speed: 0.09
- Jam Factor: 0.12

IGN scores, calculated as:

$$\text{IGN Score} = \frac{\text{LSTM Model MAE}}{\text{Climatology Baseline MAE}}$$

The LSTM model's relative improvement over static baselines highlight its ability to account for real-time variations and dynamic conditions, which are often overlooked in static baselines. By adapting to live data, the LSTM model provides actionable insights, enhancing both predictive accuracy and practical applicability. For instance, an IGN score of 0.09 for speed indicates that the LSTM model's prediction error is only 9% of the climatology baseline error. Similarly, the Jam Factor IGN score of 0.12 demonstrates significant error reduction in congestion predictions, showcasing the model's predictive strength.

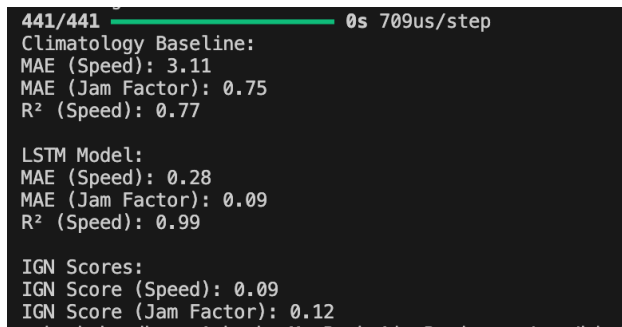


Figure 7: IGN Scores for Speed and Jam Factor

This visualization highlights how much better the LSTM model performs compared to the baseline. IGN scores show that the model's predictions are significantly more accurate for both traffic speed and congestion levels, reinforcing the model's effectiveness.

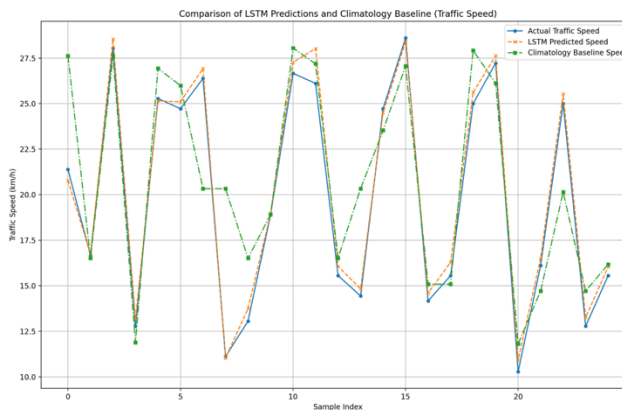


Figure 9: LSTM vs. Climatology: Speed Predictions Over Time

This line chart compares the model's predictions with the baseline over time. It highlights the LSTM model's ability to adapt to changes in traffic conditions, unlike the static climatology baseline.

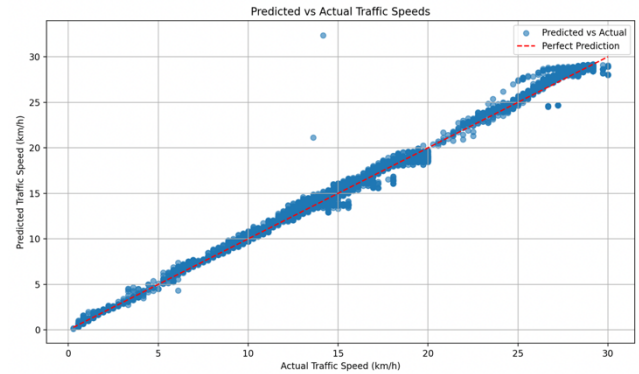


Figure 8: Comparison of Predicted and Actual Traffic Speeds

This scatter plot compares the model's predictions with actual traffic speeds. Points that align closely with the diagonal line demonstrate the model's accuracy, showing that it closely matches real-world data.

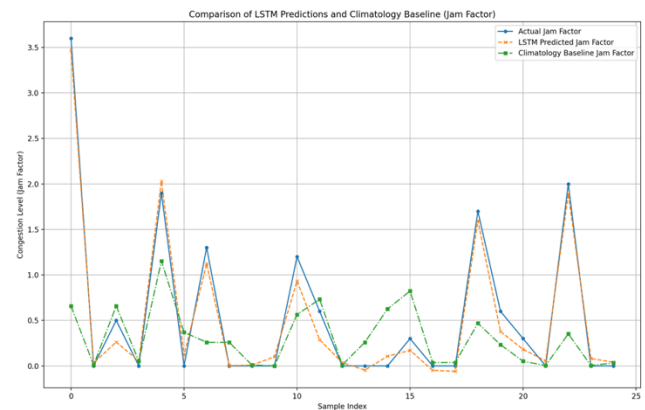


Figure 10: LSTM vs. Climatology: Jam Factor Predictions Over Time

Like Figure 9, this chart compares predictions for congestion levels. It shows how the LSTM model outperforms the baseline in capturing dynamic traffic patterns.

The following figure illustrates an example of predicted traffic speed and congestion levels for a specific scenario. It ties everything together, showcasing the model's practical applications:

```
Enter the hour of the day (0-23): 11
Enter the day of the week (1-7, where 1=Monday, 7=Sunday): 7
Available Streets:
1. Ellett Rd
2. Patrick Henry Dr
3. Roanoke St
4. Toms Creek Rd
5. US-11/US-460/Exit 118
6. US-460
7. US-460-BR/Main St
8. US-460-BR/N Main St/Farmingdale Ln
9. VA-114/Peppers Ferry Rd
10. VA-114/Peppers Ferry Rd NE
11. VA-412/Prices Fork Rd
12. VA-42/Blue Grass TR/Spruce Run Rd
13. nan
Select a street by entering the number (1-13): 4
1/1 0s 167ms/step
Predicted Traffic Speed: 8.23 km/h
Predicted Congestion Level (Jam Factor): 3.13
```

Figure 11: Final output of the prediction by the model

Insights:

The LSTM model outperforms the climatology baseline in adapting to real-time traffic variations like rush-hour congestion and weather changes. Key findings include:

- **Dynamic Adaptability:** Processes real-time inputs such as weather and temporal patterns to predict sudden traffic changes effectively.
- **Improved Accuracy:** With an IGN score of 0.09 for speed and 0.12 for jam factor, the model shows lower prediction errors than the baseline, enhancing traffic speed and congestion forecasting.
- **Practical Utility:** Demonstrates the potential of machine learning to revolutionize traffic forecasting with scalability and relevance across diverse urban scenarios.

6. Conclusion

This project demonstrates the effectiveness of LSTM networks in forecasting traffic conditions with high precision by leveraging advanced sequential modeling capabilities. The integration of real-time traffic and weather data ensures that predictions are not only accurate but also practically relevant, addressing the complexities of urban mobility challenges. The framework's adaptability and scalability ensure its relevance for diverse urban settings.

7. Future Work

The framework can be enhanced by incorporating user behavior for dynamic feedback, scaling to diverse metropolitan traffic conditions, and automating real-time data integration with continuous model updates. Additionally, analyzing the impact of public access to traffic predictions on route choices and congestion patterns can further optimize traffic management.